Solving Problems and Designing Solutions: Thomas A. Edison

Chapter 7

Thomas Alva Edison changed the world. In his lifetime he designed over one thousand inventions. Edison applied his knowledge of energy transfer and transformation to make an astounding number of useful devices.

Big Question

How did Thomas Edison use his knowledge of energy transfer and transformation to solve problems?

Edison knew a lot about the transformation of one form of energy to another. He used this expertise to develop an electric light bulb. His bulb, the carbon filament light bulb, forever changed the way people live.

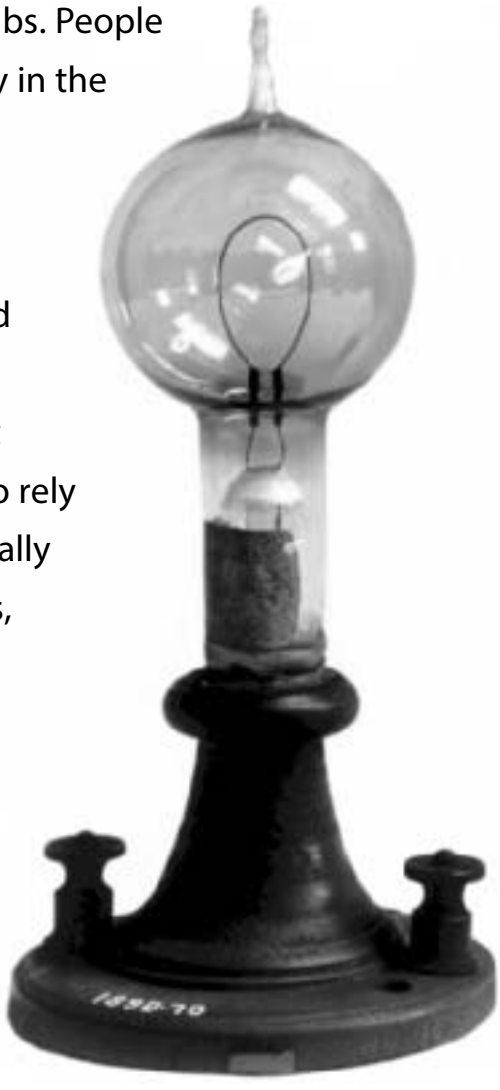
Edison designed and tested ways to use energy to solve problems. He knew that electrical energy could be transferred through wires. He also knew that, in wires made of some materials, electrical energy would be transformed to light and heat, making the wires glow brightly. Edison wanted to make a light source that would stay lit for a useful amount of time. Edison also wanted his light source to be safe and practical for people to use at home.



Born in Ohio in 1847, Thomas Alva Edison was a busy, curious boy. When he was seven, he left school and was homeschooled by his mother.

Edison Was a Problem Solver

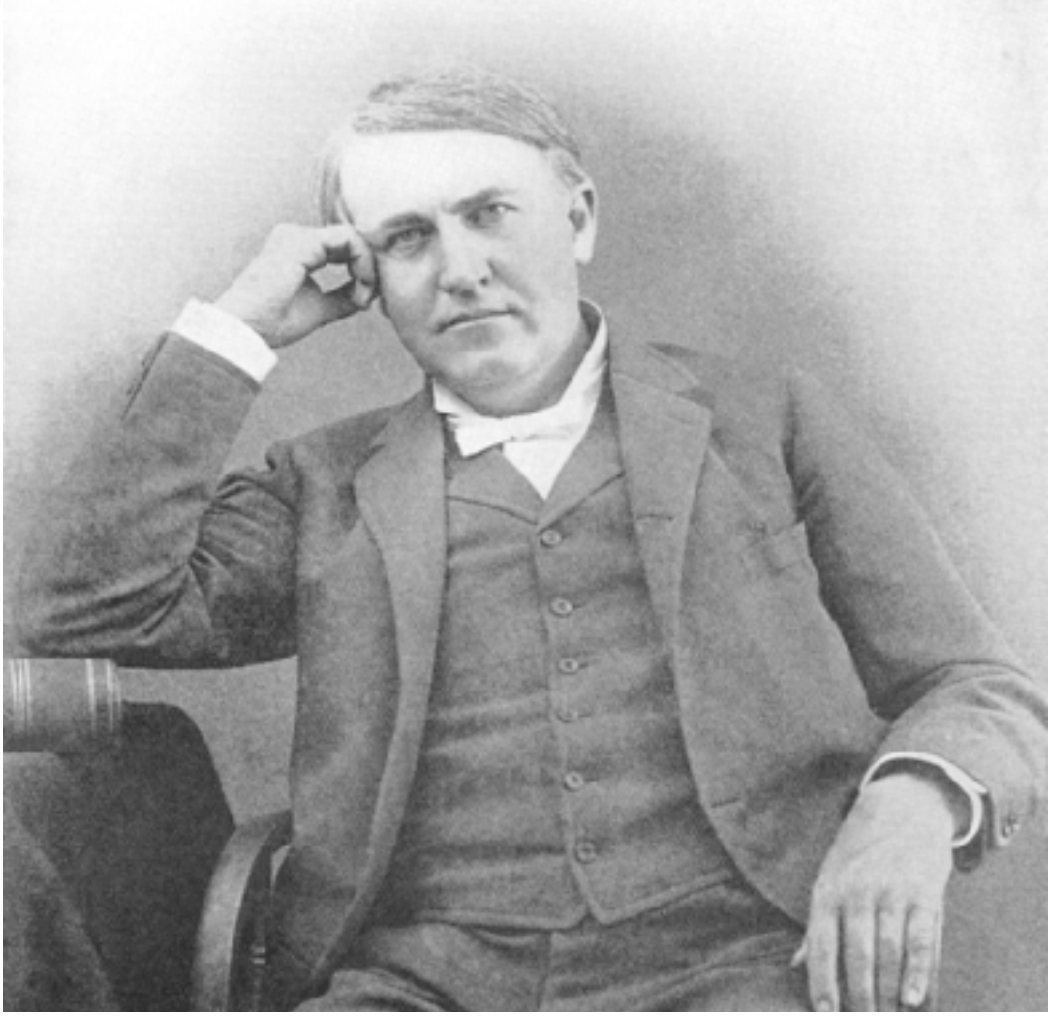
Think about a time before light bulbs. People had to rely on fire to light their way in the dark. Open flames such as torches, candles, and oil lanterns provided the light that people used for any activity after sunset. Edison worked to design a device that could use electricity to be a new kind of light source. Many people didn't want to rely on an open flame that was potentially unsafe. Eventually, after many tests, Edison found a solution.



Edison's carbon filament light bulb—the filament is the wire in a bulb that glows with light when electricity moves through it. The glass protects the filament.

Thomas Edison did not invent the first light bulb. Other scientists and engineers had designed and tested bulbs before Edison. The earliest designs of light bulbs could only provide light for a short amount of time, and the materials were expensive. Edison saw an opportunity to make the early bulbs better and more affordable for homes and businesses. Here was a great need. Thomas Edison improved the designs of others to make his famous bulb.

Edison and his team tested more than six thousand materials to find the right material for light bulb filaments. They wanted to find an inexpensive material that would glow for a long time without burning up. In 1879 he finally found a practical solution: a filament of bamboo with a coating of carbon that transformed electrical energy to light energy.



"When you have exhausted all the possibilities, remember this: you haven't."

Edison thought of engineering designs and then built, tested, and improved what he designed. He often had to conduct many tests before he found a solution that met his goal.

Electrical Energy Where You Need It

Edison's use of energy transfer and transformation extended beyond light bulbs. In 1901 he presented the first alkaline battery. Car batteries at the time used lead chemicals to store energy. Edison wanted to make batteries that would be lighter and more powerful than those heavy lead batteries. He wanted to make electric cars that would be more efficient and reliable. After many experiments and failures, he finally found a combination of chemicals that worked best.

Unfortunately, Edison's idea for electric cars didn't work out at the time. Automobiles designed to burn fossil fuels became more popular. But Edison's work with batteries often still affects your life today. Edison's alkaline battery design was used to develop the batteries you use in flashlights and remote controls.



Materials inside a battery react and transform stored energy to electrical energy. Don't break open a battery! The chemicals are toxic.

These batteries use a reaction between two chemicals to convert stored chemical energy to electricity needed in many devices. Alkaline batteries have long-lasting energy, which makes them dependable. Many modern batteries are the direct descendants of Edison's invention.

Improving Designs Never Ends

Edison famously said, "I haven't failed. I've just found ten thousand ways that won't work." Good inventors, including Thomas Edison, practice persistence. To persist means to keep trying to solve a problem, even when you find ways that won't work. Just as Thomas Edison improved the inventions of others, other inventors built upon his work.

The earliest light bulbs designed by Edison were fragile. Sometimes wires would break off from the filament. Many times, the filament would not last long because of the heat.

Lewis Howard Latimer lived in the same time period as Edison and even worked for Edison at one point. Among Latimer's seven patents are two that improve on the original light bulb design. One of his patents was a design to improve on the way the wire in a bulb was secured to the filament. Another of his patents improved the strength and durability of the filament. This meant the bulb could last longer before the filament broke from the heat.



Lewis Howard Latimer was born in 1848 in Chelsea, Massachusetts. He first worked for Edison's competitors and later for Edison's own company.

Design Solutions over Time

Many kinds of light bulbs waste a lot of the electrical energy that transfers through their filaments. This is because much of the energy changes to heat instead of just light. This is another design problem for engineers to solve. The heat weakens the bulb filament, shortening the time it lasts. The heat also warms the rooms in which people use the bulbs, which isn't welcome in spaces that people try to keep cool.

There are many types of light bulbs today that transform energy a little differently from the way Edison's bulbs worked. Fluorescent and LED light sources use less electricity, produce less heat, and last longer than other bulbs.



Traditional bulbs use electrical energy through a filament to produce light. This produces bright light and a lot of heat.



CFL (compact fluorescent light) bulbs use the same technology as neon lights. They give off a glow and little heat instead of bright light.



Tiny LED (light-emitting diode) bulbs are combined in bigger devices. They give off a lot of light and very little heat, using little energy.

Each new design improves an earlier version of the solution—providing light in a better way. In the same way, each version of the battery improves on earlier ways to provide electrical energy. As long as there are problems, people will continue to design solutions.